LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Conduits types, and insulation of wires in aircraft.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Conduits types, and insulation of wires in aircraft.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of wiring routing standards and insulation types; Instructor supervision on safety protocols.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of aircraft wiring degradation?
(b) List the common types of conduits used in aircraft wiring.
(c) Why is proper insulation critical for aircraft electrical systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of Aircraft Conduits • Rigid vs Flexible Conduit Applications • Conduit Installation Standards • Conduit Sizing and Routing • Conduit Termination Methods	L & Ds	PPT, IB	15 min
(b)	• Wire Insulation Materials • Insulation Degradation Mechanisms • Environmental Factors Affecting Insulation • Insulation Repair Techniques • Selection of Insulation Types	L & Ds	PPT, PDF	20 min
(c)	• Integrated Conduit and Insulation Systems • Maintenance Inspection Procedures • Preventive Measures for Degradation • Documentation and Reporting • Safety Considerations in Wiring	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how environmental factors contribute to wiring degradation.
(b) Differentiate between rigid and flexible conduit applications in aircraft.
(c) Describe the steps involved in inspecting conduit and insulation integrity.

12. Home Assignment and Follow Up	Prepare a brief report on the differences between conduit types and their specific applications in aircraft wiring, referencing the next lesson plan on practical repair techniques.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Practical on aircraft wiring routing and repair like crimping, insulating using heat sleeve, Solder sleeve etc.	Refer to the master syllabus reference list	LP 21

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min